



Multi-Agent System for Math Learning:

Contextualized Mathematics Multiple-Choice Question Generation with Agentic Workflow

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Abstract

In recent years, large language model-driven Multi-Agent Systems have emerged as powerful tools in education, offering innovative solutions to enhance learning experiences. The application of mathematics to solve real-world problems plays a crucial role in mathematics education. However, generating high-quality multiple-choice questions with authentic contexts presents a significant challenge. This work introduces a novel method for automatically generating contextualized multiple-choice mathematics questions by integrating multiple iterations of large language model dialogues with auxiliary external tools and the LangChain framework.

The proposed method transitions from a Robotic Workflow to an Agentic Workflow, harnessing the capabilities of multiple agents to collaborate and refine the question generation process. To evaluate the quality of the generated questions, 30 questions were randomly selected, and 13 human experts were invited to rate them. The survey results indicate that the questions produced by the proposed method exhibit significantly higher quality compared to those generated directly by GPT-4 and are already quite comparable in performance to questions meticulously crafted by humans across multiple dimensions. The implementation of an Agentic Workflow within a Multi-Agent System holds substantial potential for the future of education, offering a scalable and efficient method for creating high-quality, authentic learning materials and driving further innovations in AI-driven educational tools.

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